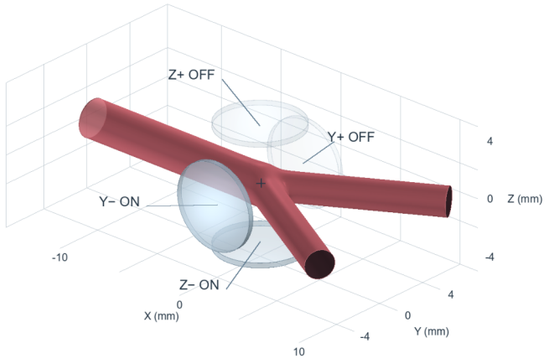
PZT shifts the cluster; robust shape control is not established

Same mature cluster, two PZT settings; actual internal rearrangements are retained

PZT layout and vessel context



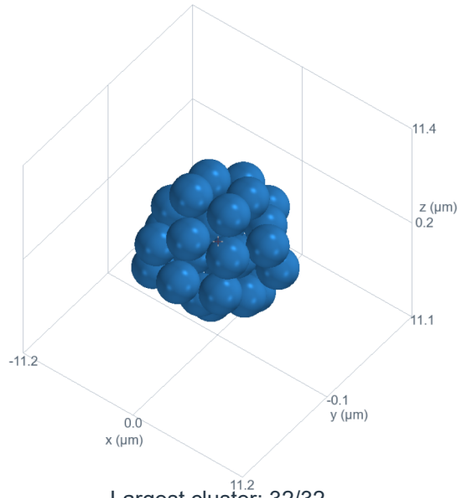
Four discs: 6 mm diameter
Centres: Y = ±4 mm, Z = ±4 mm
Cross: observation location

Local panels share a fixed camera and physical scale.

Shown: controlled PZT setting.
Unchanged: all four elements ON.

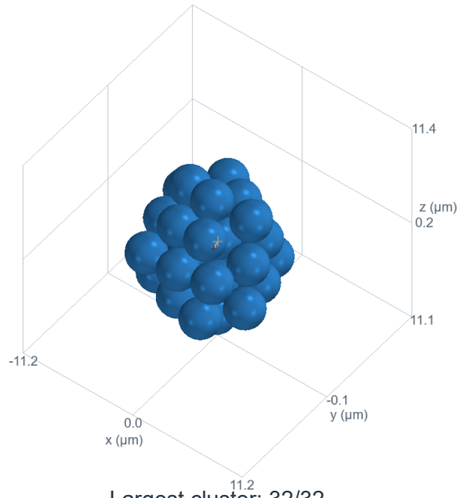
Local cross: cluster centroid.
Grey cross: initial centroid.

Same mature initial state



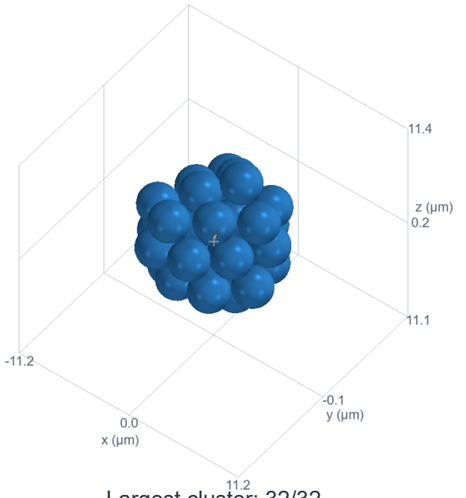
Largest cluster: 32/32

Unchanged PZT · 120 ms

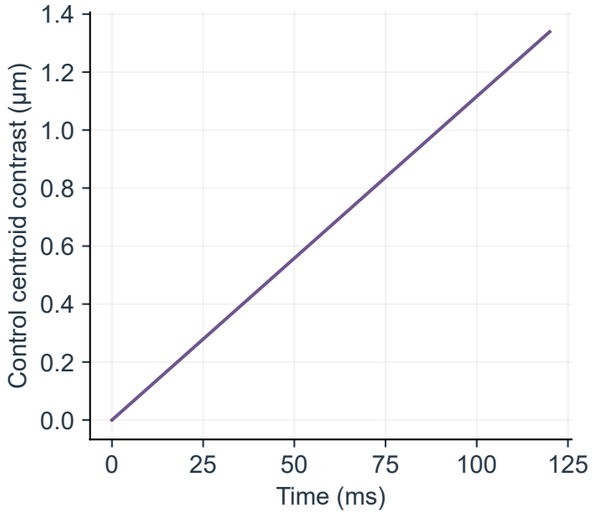


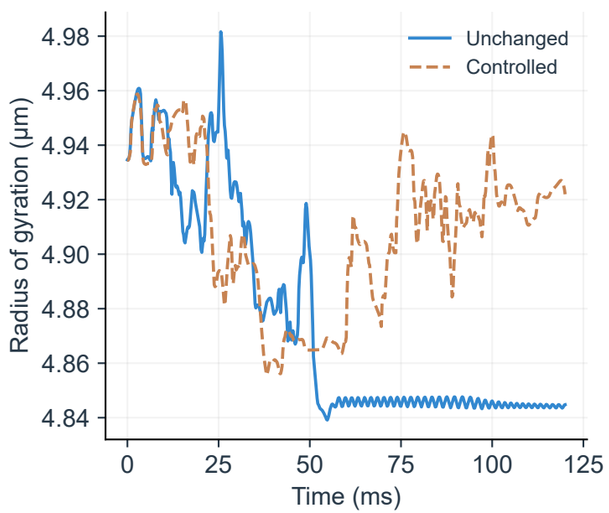
Largest cluster: 32/32

Controlled PZT · 120 ms



Largest cluster: 32/32





Translation contrast: 1.34 μm
Rg contrast: 1.59%

No sustained shape response beyond the existing sensitivity comparison was established.

Transient changes and internal rearrangements remain visible.

F3 actual trajectories • R = 1.78 μm • Zero flow; no wall forces (global vessel is context) • EXPLORATORY